

Micro-3D Printing of Packaging Substrates with Embedded Through Holes for Fan-Out Interposers

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Novelty of work

The novelty of the paper lies in its innovative approach to addressing challenges in advanced semiconductor packaging, utilizing 3D printing technology to fabricate substrates with embedded through-holes, eliminating the need for through-hole etching processes. This approach represents a departure from traditional interconnect process to build through-silicon vias, redistribution layers.

Abstract Text

In the realm of advanced semiconductor packaging, the demand for high-performance chips has escalated, giving rise to substantial interest in heterogeneous integration. High bandwidth and multifunctionality have emerged as crucial objectives for advanced packaging solutions. Traditionally, a significant portion of the cost in advanced packaging technologies stems from interconnect processes like through-silicon vias (TSVs), redistribution layers (RDLs), and Cu pillars. Even though the conventional interconnects such as TSVs, RDLs, and Cu bumps have successfully enabled chip-to-chip and chip-to-substrate connections, they come with intricate and costly processing steps, including photolithography and etching, which significantly increase the complexity and cost of the manufacturing process. In response, this paper presents a novel approach to address these challenges by utilizing 3D printing technology to fabricate substrates with embedded through-holes, eliminating the need for through-hole etching processes. This study providing the characterization of the 3D-printed substrates and the metallization process of the through holes, thereby demonstrating the feasibility.

The experimental procedure involved the design and fabrication of 5-by-5-mm-sized, one-mm-thick polymer substrates with 20- μ m-diameter embedded through holes using a 2- μ m Projection Micro Stereolithography (P μ SL) 3D printing technique. A Cu seed layer of less than 1- μ m thickness was first deposited using electroless Cu plating, followed by a pulse-reverse pulse plating process to achieve high-aspect-ratio through-hole Cu plating. The substrate's properties, including coefficient of thermal expansion, dielectric constant, and tangential loss (up to 10GHz), were analyzed. The quality of the interconnects was assessed through focused ion beam scanning electron microscopy analysis.

The use of 3D printing technology to fabricate interconnects without etching processes holds the promise of opening new avenues for advanced packaging technologies. As the demand for high-performance chips, such as AI computing and cloud computing, continues to rise, heterogeneous integration remains a significant area of interest. High bandwidth and multifunctionality are critical solutions to meet the demands of these high-performance chips, and achieving chip-to-

chip and chip-to-substrate connections with higher I/O density and shorter interconnects is central to advanced packaging. This research demonstrates the production of substrates with embedded through holes as small as 20 μm , providing a cost-effective and simplified approach to interconnect fabrication, circumventing the complexities and costs associated with traditional processes like TSV and RDL.

References

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